CalibGuard at ArGuard Shared Tasks: Threshold Calibration and Label-Space Constraints for Arabic Multimodal Hate Speech and Harmful Prompt Detection

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Abstract

This paper describes CalibGuard, our system for the ArGuard 2026 Shared Task on harmful content detection in Arabic memes and large language model prompts. We participate in all four subtasks: binary and fine-grained hateful meme classification (A1, A2) and binary and harm-domain prompt classification (B1, B2). For Task A, we extract frozen embeddings from three vision-text encoder pairs and train a joint two-headed classifier under 5-fold cross-validation with seed averaging. For Task B, we fine-tune AraBERTv02 with class-weighted cross-entropy. CalibGuard obtains 75.53% macro-F1 on subtask A1, 36.53% on A2, 65.38% on B1 and 68.85% on B2, ranking 11th, 3rd, 15th and 13th respectively. We further show that the B1 test split contains 49.3% unsafe prompts against 79.8% in training, and that this distribution shift accounts for a 29-point gap between our development and test scores, concentrated entirely in Modern Standard Arabic. Rethresholding to the test prior recovers 13.7 points without retraining. We additionally report two negative results: constrained decoding marginally reduces A2 performance, and restricting B2 training to unsafe prompts costs 5.7 points.

1 Introduction

Harmful content in Arabic propagates through two channels usually studied in isolation. Memes convey hate through the interaction between image and overlaid text, so neither modality is reliable alone; prompts directed at large language models convey harm through intent, across a dialect continuum. The ArGuard 2026 Shared Task (Alam et al., 2026) addresses both, using AHA-Memes (Kmainasi et al., 2026) for Task A and a dialect-annotated prompt corpus for Task B, evaluated by macro-F1.

Three properties of this task present difficulties a conventional fine-tuning pipeline does not address. Task A releases only 4,500 labelled memes, at which scale large vision-language models overfit readily. Its training and evaluation splits differ in class prior and annotation protocol, so the labels a system fits are noisier than those against which it is scored. And the Task B training split diverges substantially from the evaluation split in label distribution, a form of label shift (Quiñonero-Candela et al., 2009) penalising any threshold estimated on in-distribution data.

CalibGuard addresses these difficulties rather than model capacity: we freeze the encoders and place all trainable parameters in a small head; select thresholds on out-of-fold predictions spanning the entire labelled pool; and correct explicitly for the documented prior shift. Our contributions are:

- a calibration-oriented multimodal system ranking 3rd of 11 on subtask A2 without fine-tuning any vision-language model, exceeding the organiser baseline on all four subtasks;
- a quantification of label shift in subtask B1, which accounts for a 29-point development-to-test gap and of which rethresholding recovers 13.7 points;
- evidence that this degradation is confined to Modern Standard Arabic (MSA), inverting the difficulty ordering suggested by development data, with two negative ablation results.

2 Related Work

Multimodal hate detection. The Hateful Memes Challenge (Kiela et al., 2020) showed that unimodal shortcuts score deceptively well and that benign confounders force cross-modal reasoning; later work extended this to persuasion techniques (Dimitrov et al., 2024) and Arabic propagandistic memes (Alam et al., 2024). ArAIEval 2024 (Hasanain et al., 2024) is the closest predecessor to Task A, its stronger systems pairing an Arabic text encoder with a vision backbone through a com-

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pact fusion module (Wang and Markov, 2024); we follow this design but freeze both encoders.

Arabic encoders and dialect. Model choice interacts with dialectal variation: MARBERT (Abdul-Mageed et al., 2021) is pre-trained on tweets and covers dialectal forms, whereas AraBERT (Antoun et al., 2020) and CAMeLBERT (Inoue et al., 2021) weight Modern Standard Arabic more heavily; prior work on Arabic offensive language exploits these differences (Mubarak et al., 2023; Alam et al., 2022). Safety-specific resources for Arabic language models are more recent, comprising general (Mubarak et al., 2025) and dialect-aware (Zaghouani et al., 2026) benchmarks. Since the ArGuard corpus spans six varieties, we treat encoder diversity as a source of ensemble variance.

Distribution shift. The failure mode we document is label shift (Quiñonero-Candela et al., 2009): $P(y)$ differs between training and test while $P(x \mid y)$ is approximately stable. Classical procedures adjust outputs to a new prior (Saerens et al., 2002), black-box estimators recover it from unlabelled data (Lipton et al., 2018), and macro-F1 thresholding is sensitive to calibration (Guo et al., 2017). Our contribution here is empirical: this failure mode is present, severe and undocumented in a live shared task, and interacts with dialect in a manner aggregate scores conceal.

3 Data

Task A. AHA-Memes (Kmainasi et al., 2026) provides labelled Train, Dev and Dev_test splits and a blind Test split; each sample comprises an image, machine-extracted text, a binary label and fine-grained labels. Since all three non-blind splits are labelled, we pool them into 4,500 samples, 28% more than Train alone (Table 1). Two properties inform our design: the binary prior differs between the pool (37.8% hateful) and Test (29.6%), and the training splits are single-annotator whereas Test is triple-annotated. Five of fifteen fine-grained categories lack training support and are ignored by the scorer, leaving ten active labels, of which `Exclusion` (n=14) and `Inferiority` (n=80) are rare.

Task B. The prompt corpus (Zaghouani et al., 2026) contains 16,835 training and 2,405 development samples, each with a dialect tag, a binary label and a harm domain. We remove an unnamed index column from the released test files, discard 4 samples with empty prompts or missing labels, and normalise whitespace, retaining 16,831 samples. Three characteristics affect interpretation. The Dev split is optimistic: 11 prompts appear verbatim in training and around 10% lie within 0.8 character n-gram cosine similarity of a training sample, so we hold out an internal split of 2,020 samples stratified by label, harm domain and dialect. The two test files differ in size (7,809 for B1, 5,979 for B2) and are separate samples. Harm domains are assigned to safe prompts too, so 37% of `harassment` samples carry a safe label; we restrict B2 to unsafe prompts (11,823), inferring this from the test file size ratio (0.766) against the training unsafe rate (0.798), which Section 5 shows was incorrect.

Task A (memes)				
Split	Train	Dev	Dev_test	Test
# Samples	3,500	500	500	500
% Hateful	37.8 (pool n=4,500)			29.6
Annotation	Single			Triple

Task B (prompts), % by variety		
	Train (16,835)	Test B1 (7,809)
Modern Standard	48.0	68.3
Syrian	21.7	13.2
Moroccan	10.0	6.0
Egyptian	9.5	5.6
Palestinian	6.6	4.5
Algerian	4.1	2.5
% Unsafe	**79.8**	**49.3**

Table 1: Dataset statistics. Both tasks exhibit shift between training and evaluation: Task A in class prior and annotation protocol, Task B in dialect composition and label prior.

4 Methodology

All experiments use a single Kaggle T4 GPU with PyTorch, Transformers (Wolf et al., 2020) and scikit-learn (Pedregosa et al., 2011), in under one GPU-hour total.

4.1 Task A: Frozen Encoders with a Joint Head

Given 4,500 labelled samples, fine-tuning a large vision-language model invites overfitting, so we freeze the encoders and allocate all trainable capacity to a compact head (Figure 1). Representations are extracted once from three vision-text pairs and cached:

- **SigLIP** (Zhai et al., 2023) with **MARBERTv2** (Abdul-Mageed et al., 2021): a sigmoid-loss image encoder with the most dialect-aware Arabic encoder.

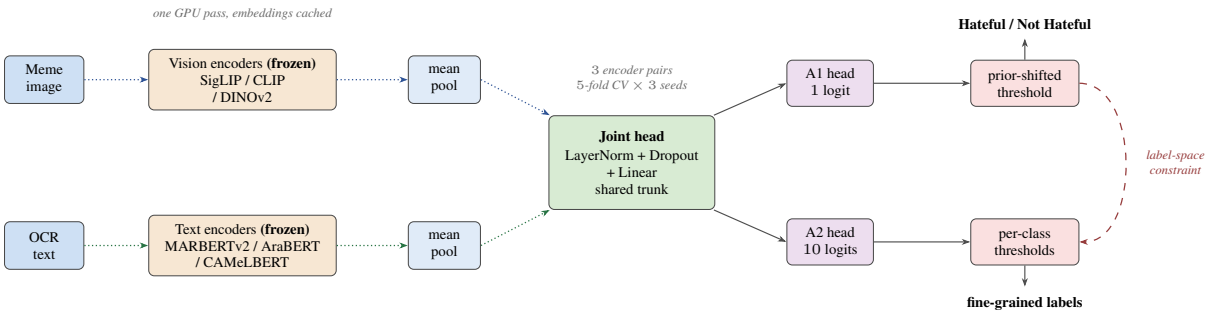


Figure 1: The CalibGuard architecture for Task A. Both encoders remain frozen and their embeddings are cached, so the joint head is the only trained component. Dotted arrows mark the visual and textual paths. The A1 decision is applied as a mask over the A2 label space, since hateful and non-hateful sub-types never co-occur.

- **CLIP** ViT-B/32 (Radford et al., 2021) with **AraBERTv02-twitter** (Antoun et al., 2020), whose social media pre-training suits meme text.
- **DINOv2** (Oquab et al., 2024) with **CAMeLBERT-mix** (Inoue et al., 2021): a self-supervised backbone with an encoder covering standard, dialectal and classical Arabic.

Images are mean-pooled over patch tokens and text over subword tokens with attention masking. Each pair feeds a joint head of two LayerNorm-Dropout-Linear projections, a shared trunk and two outputs: one logit for A1 and ten multi-label logits for A2, trained jointly with class-weighted binary cross-entropy for 60 epochs at learning rate 10^{-3}, batch size 128, dropout 0.4, and label smoothing ϵ=0.05 (Szegedy et al., 2016). We use 5-fold stratified cross-validation with three seeds per fold, stratifying on the binary label crossed with the dominant fine-grained label; each fold model predicts its held-out partition and the blind Test split, averaged over 45 models. Ensembling standardises logits using out-of-fold statistics applied to the test logits, since standardising each split independently would remove the prior shift we correct.

Calibration. Thresholds are selected on the 4,500 out-of-fold predictions rather than the 500-sample Dev split, since macro-F1 is sensitive to the operating point and a threshold estimated on 500 samples overfits (Guo et al., 2017). A1 uses a single threshold and A2 one per class. We prepare two submissions, at the out-of-fold optimum and rate-matched to the documented 29.6% test prior.

Constrained decoding. A hateful meme never carries `Humor` or `Sarcasm`, and a non-hateful meme never carries a hateful sub-type, while `Other` is shared. Conditioned on the A1 prediction we mask the disallowed block before thresholding, falling back to the arg-max of the permitted set when no label passes.

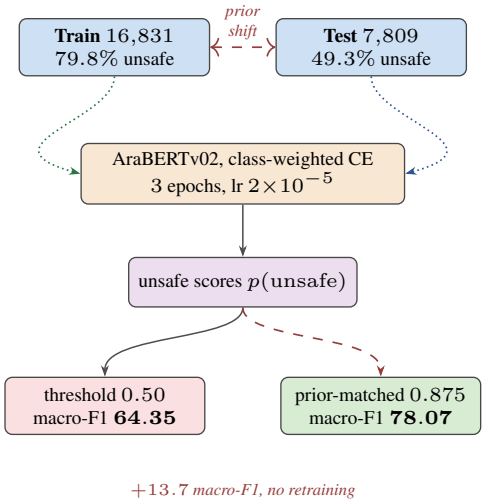


Figure 2: The Task B pipeline. Training and evaluation carry different unsafe priors, so a model fitted to the training ratio is miscalibrated at the default threshold; matching the operating point to the test prior recovers 13.7 macro-F1 without retraining (Section 6).

4.2 Task B: Fine-tuned Arabic Encoder

We fine-tune AraBERTv02 (Antoun et al., 2020) with class-weighted cross-entropy, truncating prompts to 128 tokens, for three epochs at learning rate 2×10^{-5} with batch size 32. Class weighting is necessary since the training split is 79.8% unsafe and macro-F1 penalises neglect of the minority class. A character n-gram TF-IDF and logistic regression model serves as a reference floor that any neural configuration must exceed. B1 and B2 are trained separately; we use no external data,

System	macro-F1
Subtask A1, out-of-fold	
SigLIP + MARBERTv2	69.38
DINOv2 + CAMeLBERT-mix	68.69
CLIP + AraBERT-twitter	67.93
Ensemble	**71.18**
Official test (rank / teams)	
A1 Ensemble	75.53 (11/16)
A2 Ensemble	**36.53 (3/11)**
B1 AraBERTv02	65.38 (15/20)
B2 AraBERTv02	68.85 (13/16)
Baseline A1 / A2	67.80 / 29.50
Baseline B1 / B2	59.20 / 68.70

Table 2: Out-of-fold and official test results (%). The A1 ensemble exceeds its best member by 1.8 points, and CalibGuard clears the organiser baseline on all four subtasks. Development scores are given in the text.

although AraSafe (Mubarak et al., 2025) was permitted.

5 Results

Table 2 reports out-of-fold and official test results. CalibGuard exceeds the organiser baseline on all four subtasks; its best placement is A2, 3rd of 11 and 7.0 points above baseline.

Task A. SigLIP with MARBERTv2 is the strongest individual pair and the ensemble exceeds it by 1.8 points, indicating the members produce different errors. Per-class A2 performance tracks support: the rarest labels attain 2.5 (`Exclusion`) and 14.8 (`Inferiority`) out-of-fold F1, whereas the four best-supported exceed 46.

Task B. On development, the character n-gram reference model scores 91.22 on B1 and 85.52 on B2, while AraBERTv02 reaches 94.41 and 87.82, exceeding the floor by 3.2 and 2.3 points; its predictions constitute our submission.

Ablations. Two design decisions are not supported by measurement. Constrained decoding, expected to benefit A2, marginally reduces out-of-fold macro-F1 from 34.51 to 34.13, yet was applied to the submitted run, placing our A2 result about 0.4 points below the unmasked decoder. Restricting B2 training to unsafe prompts is costlier: training on all 16,831 samples attains 71.53 against 65.85, since safe-labelled samples carry a harm domain.

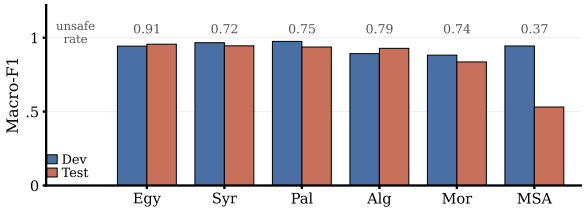


Figure 3: Subtask B1 macro-F1 by dialect. The figure above each group is that dialect's true unsafe rate on test; performance degrades with distance from the 0.798 training prior.

6 Error Analysis

6.1 Label Shift in Subtask B1

Our B1 system attains 94.41 macro-F1 on development and 65.38 on test. The training and development splits are 79.8% and 80.2% unsafe whereas the B1 test split is 49.3% (Table 1), a shift undocumented in the task description: a model trained under a 4:1 ratio is calibrated for a distribution the evaluation data does not exhibit, and our predictions were 78.7% unsafe. The cause is the data rather than the architecture, since the reference model also drops from 91.84 to 64.35. Rethresholding recovers most of the loss without retraining: the reference model scores 64.35 untuned, 57.46 tuned on development and 61.24 out-of-fold, but 78.07 when matched to the test prior, within 0.1 of the 78.18 oracle. Out-of-fold tuning beats development tuning by 3.8 points, yet both trail the untuned threshold, since every in-distribution estimate optimises for the source prior.

6.2 Degradation Confined to Modern Standard Arabic

Aggregate scores conceal where the loss occurs. Figure 3 decomposes B1 macro-F1 by dialect: every dialect is stable on test, four of five above 92 and Moroccan at 83.6, whereas Modern Standard Arabic falls to 53.1, with accuracy tracking each dialect's distance from the training prior. Modern Standard Arabic is 36.6% unsafe on test while every dialect stays between 71.7% and 91.3%, and its share rises from 48.0% to 68.3%, so the test set appears to have been rebalanced with safe Modern Standard Arabic prompts. This inverts the development ordering, where Moroccan (88.2) and Algerian (89.2) were weakest and Modern Standard Arabic appeared robust at 94.4.

A lexically separable harm domain. The `adult_content` class is separable to a degree

that does not reflect the difficulty of the problem: the reference model attains 100 precision and recall on development and 99.3 on test. These prompts average 6.9 words against 11 to 13 elsewhere and 493 two-word prefixes cover 5,249 samples, which we attribute to corpus construction. At 21.9% of the B2 test split it inflates macro-F1, which falls to 83.11 when excluded.

7 Conclusion

We presented CalibGuard, a calibration-oriented system entered in all four ArGuard 2026 subtasks, which ranked 3rd of 11 on fine-grained meme classification while fine-tuning no vision-language model. Freezing the encoders and confining trainable capacity to a joint two-headed classifier proved an effective response to a 4,500-sample training set, and ensembling three encoder pairs over cross-validated folds added 1.8 macro-F1 points out-of-fold.

Our most transferable result, however, concerns a failure rather than a gain. The Task B evaluation split carries a label prior substantially different from its training data, an undocumented shift that costs 14 macro-F1 points and that a post-hoc adjustment of the decision threshold recovers without retraining. Two further results ran against our expectations: threshold selection on in-distribution data proved worse than no selection at all under this shift, and restricting B2 training to unsafe prompts discarded samples that remain informative for harm-domain boundaries. Together with the finding that the degradation is concentrated entirely in Modern Standard Arabic while every regional variety remains stable, these results suggest that systems for Arabic content safety would benefit from estimating the evaluation prior from unlabelled test data and from reporting macro-F1 per dialect rather than in aggregate.

Limitations

Our Task A ablations are measured on out-of-fold predictions rather than on the test split, so the contributions of individual ensemble members, of label smoothing and of constrained decoding are established under cross-validation rather than under the evaluation distribution. A test-set replication of these ablations, together with an error analysis of subtask A1 comparable in depth to our Task B analysis, remains future work.

The prior-matched threshold uses the true test prior, which is unavailable to participants during evaluation. It establishes the magnitude of the recoverable gap rather than a deployable procedure; obtaining it in practice requires estimating the target prior from unlabelled data (Lipton et al., 2018), and such estimates carry their own error. Finally, our dialect conclusions rest on the tags distributed with the corpus, which we did not independently verify, and the two Maghrebi varieties have the smallest test subsets (193 and 469 samples), making those estimates the least stable.

Ethical Considerations

This work involves models trained on offensive material, including hate speech, harassment and prompts describing self-harm. The data was used solely for the shared task, under its licence, and was not redistributed; the meme corpus carries a non-commercial licence and our working copies were kept private. No examples of harmful content are reproduced in this paper, and our discussion of the `adult_content` artifact reports only aggregate statistics rather than surface patterns that would assist in evading a filter.

Two risks merit emphasis. Our per-dialect results demonstrate that a single aggregate score can conceal a 40-point spread across varieties of the same language, so that a classifier deployed on the basis of an aggregate figure may fail disproportionately for speakers of particular dialects, which constitutes a fairness concern rather than solely an accuracy one. Separately, the calibration result applies in both directions: a system whose operating point is tuned to an incorrect base rate over-flags safe content, which for a safety filter entails suppressing benign speech. Neither our systems nor the underlying benchmark should be regarded as deployment-ready.

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